MAT1100 Homework 3

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Question 3

(a) *Proof.* Consider the map $f : k[x] \to k^2$ given by:

$$f : p \mapsto (p(0), p(1))$$

this is clearly a morphism of rings. Now suppose $p \in \ker f$ this occurs if and only if $p(0) = 0$ and $p(1) = 0$ which is holds if and only if $x \mid p$ and $(x-1) \mid p$ and so

$$\ker f = \langle x(x-1) \rangle$$

now to show that f is surjective suppose $(a, b) \in k^2$ then

$$f((b-a)x + a) = (a, b)$$

thus f is surjective. Now, by the 1st isomorphism theorem, f restricts to an isomorphism

$$\overline{f} : k[x]/\langle x(x-1) \rangle \to k^2$$

given by

$$\overline{f}(\overline{p}) \to (p(0), p(1))$$

note that this implies that every class in A is uniquely represented by a polynomial of the form $ax + b$, which will be useful in part (c). □

(b) *Proof.* To find the ideals of A it suffices to classify the ideals I lying between $\langle x(x-1) \rangle$ and $k[x]$ by the ideal quotient correspondence. Since k is a field, $k[x]$ is a PID which means this question reduces to finding all monic polynomials dividing $x(x-1)$ and these are clearly $1, x, (x-1), x(x-1)$. Thus the ideals of A are $(0), (x)/(x)(x-1), (x-1)/(x)(x-1), A$. Moreover the 3rd isomorphism theorem implies

$$\frac{k[x]/(x)(x-1)}{(x)/(x)(x-1)} \cong k[x]/(x) \cong k$$

hence $(x)/(x)(x-1)$ is maximal and hence prime, and by a similar argument so is $(x-1)/(x)(x-1)$. Moreover it is clear that (0) and A are not prime or maximal. To see that (0) is not prime note that $[x][x-1] = [x(x-1)] = 0$ hence A has zero divisors and so (0) cannot be prime.

Now to find the ideals of B we apply a similar argument only to the polynomial x^2 and from this we find it has monic divisors $1, x, x^2$ corresponding to ideals of B $(0), (x)/(x^2), B$ where the middle ideal is maximal as

$$k \cong k[x]/\langle x \rangle \cong \frac{k[x]/\langle x^2 \rangle}{\langle x \rangle/\langle x^2 \rangle} \cong \frac{B}{\langle x \rangle/\langle x^2 \rangle}$$

and the other two are neither prime nor maximal. To see that (0) is not prime note that $[x][x] = [x^2] = 0$ hence B has zero divisors and so (0) cannot be prime. □

(c) *Proof.* To find the units of A we use part (a). Note that an element $(a, b) \in k^2$ is a unit if and only if $a, b \neq 0$ as then (a^{-1}, b^{-1}) is its inverse. This corresponds to polynomials of the form $(b-a)x+a$ where $a \neq 0$ and $b \neq 0$. I.e. the units of A are all polynomials classes represented by $ax + b$ where $b \neq 0$ and $a \neq b$. Now $(a, b) \in k^2$ is nilpotent if and only if a and b are which is only possible if $a = b = 0$ as k is a field. Thus the only nilpotent element in A is 0. Now the idempotents in k^2 are (a, b) such that $a^2 = a$ and $b^2 = b$ but the only idempotents of k are 0 and 1 thus the idempotents of k^2 are $(0, 0), (1, 0), (0, 1), (1, 1)$. These correspond to the polynomials $0, 1, x, -x + 1$.

To find the units of B first we write out the multiplication of two representatives $ax + b$ and $cx + d$:

$$(ax + b)(cx + d) = acx^2 + (ad + bc)x + bd = (ad + bc)x + bd \mod x^2$$

where it is clear that all classes in B can be uniquely represented by a polynomial of the form $ax + b$. Now if $ax + b$ is a polynomial for its class to be a unit in B clearly b must be a unit, i.e. $b \neq 0$. Then consider:

$$-ab^{-2}x + b^{-1}$$

clearly modulo x^2 we have

$$(ax + b)(-ab^{-2}x + b^{-1}) = (ab^{-1} - bab^{-2})x + bb^{-1} = 1$$

hence the units of B are precisely the classes of polynomials $ax + b$ with $b \neq 0$.

Now if a polynomial $ax + b$ is nilpotent it is not a unit so $b = 0$. But then

$$(ax)^2 = a^2x^2 = 0 \mod x^2$$

thus any polynomial of the form ax represents a nilpotent class in B. Finally an element of B is idempotent if and only if modulo x^2

$$ax + b = (ax + b)^2 = 2abx + b^2$$

in other words $a = 2ab$ and $b = b^2$. This implies that $b = 0, 1$ as k is a field and in either case this forces $a = 0$. Thus the only idempotents in B are 0 and 1. □

(4) *Proof.* A and B cannot be isomorphic as rings as a ring isomorphism preserves the number of idempotents in a ring, but A has 4 idempotents while B has 2. Hence the two rings cannot be isomorphic. □